

MongoDB Aggregation Examples

1. \$match

The \$match stage filters the documents to pass only the documents that match the specified condition(s).

Example:

```
db.orders.aggregate([  
  { $match: { status: "A" } }  
])
```

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2. \$group

The \$group stage groups input documents by the specified _id expression and for each distinct grouping, outputs a document.

Example:

```
db.orders.aggregate([  
  { $group: { _id: "$cust_id", total: { $sum: "$amount" } } }  
])
```

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3. \$project

The \$project stage reshapes each document in the stream, such as by adding new fields or removing existing fields.

Example:

```
db.orders.aggregate([  
  { $project: { item: 1, discount: { $subtract: ["$price", "$fee"] } } }  
])
```

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4. \$sort

The \$sort stage sorts all input documents and returns them to the pipeline in sorted order.

Example:

```
db.orders.aggregate([  
  { $sort: { amount: -1 } }  
])
```

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5. \$limit

The \$limit stage limits the number of documents passed to the next stage in the pipeline.

Example:

```
db.orders.aggregate([  
  { $limit: 5 }  
])
```

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6. \$lookup

The \$lookup stage performs a left outer join to a collection in the same database to filter in documents from the "joined" collection for processing.

Example:

```
db.orders.aggregate([  
  { $lookup: { from: "inventory", localField: "item", foreignField: "sku", as: "inventory_docs" } }  
])
```

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7. \$unwind

The \$unwind stage deconstructs an array field from the input documents to output a document for each element.

Example:

```
db.orders.aggregate([  
  { $unwind: "$items" }  
])
```

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8. \$addFields

The \$addFields stage adds new fields to documents. The stage outputs documents that contain all existing fields from the input documents and newly added fields.

Example:

```
db.orders.aggregate([  
  { $addFields: { totalPrice: { $sum: [ "$price", "$tax" ] } } }  
])
```


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9. \$replaceRoot

The \$replaceRoot stage replaces the input document with the specified document.

Example:

```
db.orders.aggregate([  
  { $replaceRoot: { newRoot: "$item" } }  
])
```

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10. \$facet

The \$facet stage processes multiple aggregation pipelines within a single stage on the same set of input documents.

Example:

```
db.orders.aggregate([
  { $facet: {
    "price": [
      { $match: { price: { $exists: 1 } } },
      { $bucket: { groupBy: "$price", boundaries: [ 0, 200, 400 ], default: "Other" } }
    ],
    "category": [
      { $match: { category: { $exists: 1 } } },
      { $bucket: { groupBy: "$category", boundaries: [ "A", "B", "C" ], default: "Other" } }
    ]
  }
}]
```